Inorganic Cume – Point Groups **October 25, 1997**

Because of the large variety of molecular structures found in inorganic chemistry, symmetry (point group) notation is often used to describe bonding geometries. For example, rather than say a three-atom fragment is either bent or linear, articles might say it was either C_{2v} or $D_{\infty h}$.

Demonstrate your mastery of this inorganic vocabulary (and a general knowledge of inorganic materials) by giving a sketch and the point group for each of the following:

1. *Cis*-dichlorotetrafluorocobaltate(III)
2. *Trans*-dichlorotetrafluorocobaltate(III)
3. *Facial*-trichlorotrifluorocobaltate(III)
4. *Meridial*-trichlorotrifluorocobaltate(III)

5.-7. Three possible isomers of PF_3Cl_2

8.-10. Three possible isomers of $B_5C_2H_7$

11. $P_3N_3F_6$
12. Iodine trifluoride
13. Xenon tetrafluoride
14. Ferrocene
15. Phosphine
16. Sulfur trioxide
17. Sulfite ion
18. NO_2^+
19. NO_2^-
20. The white phosphorus molecule